Technical, economical, and environmental optimization analysis of the hybrid energy system on Djerba Island

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***Abstract*— The objective of the research consists in providing a technical-economic, and environmental assessment of renewable energy resource options. A hybrid system consisting of photovoltaic, wind turbine, tidal turbine, and hydraulic power with battery was developed from outside the network, with diesel generators providing backup power. The design of the system has been analyzed with Hybrid Multi-Energy Resource Optimization software, which simulates and optimizes the distribution of energy based on consumption and energy sources, as well as calculating pollutant gas emissions. The financial report calculated the net present and total energy unit costs. This system has been designed to satisfy the demand for energy in south-eastern Tunisia on the island of Djerba. The suggested system is the best because it considers power generation, cost of energy, emissions, and net present cost.**

Keywords*— Economic analysis, Net present cost, Hybrid system, HOMER software.

I. Introduction

In the last few years, minimizing energy costs, and environmental pollution have become major concerns for people worldwide, at a time when consumption of fossil fuels has fallen dramatically and the rate of climatic change is accelerating in recent years [1]. With intelligent grids and new information and communication developments, greenhouse emissions are being lowered and air quality is being improved around the world [2] and in the micro-grids [3], meaning considerably reduced property costs and greater benefits for consumers of both residential and commercial properties.

In [4], an optimum dimensioning approach is described for a micro island grid in Djerba, Tunisia, consisting of the solar photovoltaic system, wind system, and tidal system. A dedicated generator and a storage battery system are used to offset this.

In [5] they propose a renewable energy hybrid distribution system using a new concept, designed to minimize both greenhouse gas emissions and costs. Different distribution network connections are used to interconnect these centers, which have different thermal and electrical distributions. Each network center's energy supply is from the output of major electrical power generation plants, from natural gas, from other energy centers on the network, or renewable resources. The analysis performed in this paper takes into account multi-objective functions to minimize costs and reduce carbon emissions.

Today, the use of renewable energy systems (HRES) has become an important part of electricity generation in the rural environment. Many reports [6] have examined several hybrid renewable energy system (HRES) designs, and have supplied crucial background information for the creation of autonomous HRESs. In [7], researchers investigated microgrid design and development on the small Tunisian island of Djerba. The HOMER software application was used to better understand the technical and economic background.

One of the main challenges of isolated systems is to synchronize energy supply with fluctuating electricity demand. This problem is solved by integrating an energy storage system (ESS), which stores excess electrical energy and then meets demand during peak periods [8]. Many researchers stress the importance of ESS, particularly for load frequency control, reliability, and system quality in isolated grids [9]. However, some studies of high-penetration renewables assume that generation should never exceed demand [10].

[11] presents a summary of renewable energy systems that have an effect of durability on the energy used. It also considers the main challenges and considerations associated with system configuration and energy supply management. In their paper [12], the authors focus on cost reduction and reduced CO_2 emission by resolving multi-objective unity of engagement issues. However cost and emissions are scaled differently, which makes it challenging to effectively link them up.

The authors used HOMER software to carry out a technical and economic evaluation of an autonomous microgrid system based on solar photovoltaic energy, wind power, and battery storage [13].

In this study, we developed both the economic and the environmental aspects of the design of various HRES (photovoltaic/wind turbine/hydro turbine/hydraulic) integrating other power systems like a diesel and a battery. To assess the suggested system, the power generation of an island, Djerba, in the southeast of Tunisia, has been taken for a case study. The innovation of this study is that, in addition to technical and economic factors, the environment is also taken into account, resulting in a solid analysis that can be used to make decisions on the design of energy solutions using hybrid systems.

In section 1, we describe the proposed hybrid renewable energy system. The section 2, we describe an economic analysis of the system. Then, we discuss the simulation results that demonstrate the efficiency of the suggested method and provide conclusions in our final section.

II. OVERVIEW SYSTEM AND BACKGROUND OF OUR STUDY

A. HOMER system

The HOMER software program, designed by the National Renewable Energy Laboratory (NREL) of the United States, is designed to provide the technical and commercial assessment required for hybrid energy systems. It can be used for modeling the physics of a production system and evaluating its life-cycle costs, including investment and operating costs. With HOMER, it is possible to analyze a wide variety of configurations based on technical and economic criteria. It can take into consideration uncertainties in input data and variations in input parameters to identify the optimum solution from both a technical and economic point of view [7].

The software's main features comprise:

- Modeling of electrical, hydrogen, and thermal generation loads.
- Simulation of photovoltaics, wind turbines, water turbines, and fossil fuel generators.
- Consideration of grid connection in different scenarios.
- Economic assessment of different types of technology.
- Assessment of the impact on the environment of different power solutions.
- Simulation and analysis of several scenarios to determine the most appropriate option for the specific requirements of the project.

B. Details case study

The area selected for the installation of a hybrid system comprising photovoltaic panels, wind turbines, tidal turbines, a hydraulic generator, and a diesel generator with batteries is situated in an isolated area on the island of Djerba, Tunisia. Its geographical situation offers abundant water sources and a lot of solar radiation.

Fig. 1. Study area location

The area studied is situated in the south-eastern part of Tunisia, on the island of Djerba. It is situated on the Gulf of Gabes and extends over an overall surface of 514 km². The island's coordinates are 33° 48′ N and 10° 51′ E. These features make the site a suitable location for the design of an HRES, as illustrated in Fig. 1.

The system suggested offers a sustainable solution on the Island of Djerba, illustrated in Fig. 2. It is composed of a solar photovoltaic panel, a wind system, a tidal system, a hydroelectric generation source, and a diesel generator. The detailed costs associated with the different components of the suggested hybrid system configuration are presented in Table. I. An analysis of the main cost items was carried out using HOMER software, including initial investment, replacement, and operation and maintenance (O&M) costs for the various energy sources. The National Aeronautics and Space Administration (NASA) power platform provided climate and energy load data for the study area, which were integrated into the analysis to refine the system modeling.

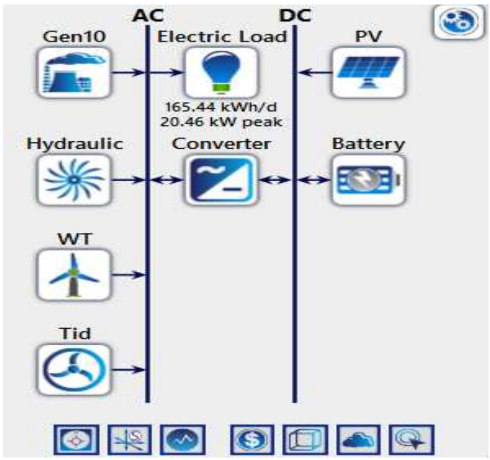

Fig. 2. HOMER simulation structure

TABLE I. COSTS OF EACH HYBRID SYSTEM COMPONENT [14]

Components	Capital cost ($)	Replacement cost ($)	O&M cost ($/year)
PV System	1500	3500	10
Wind Turbine	10 000	10 000	400
Tidal Turbine	10 000	1535	200
Hydraulic	1000	100	100

C. The environmental data

In Fig. 3 we present the yearly irradiance of the sun in Tunisia for the island of Djerba, based on the NASA site, with the lowest solar radiation seen in December at 2.69 kWh/m2/day, and irradiance was highest for July at 7.62 kWh/m2/day.

Fig. 4 illustrates the distribution of wind speeds, with the mean wind speed over the island at 5.89 m/s. The mean speed of the tides equal also 0.75 m/s, is illustrated in Fig. 5. In Fig. 6, water flow is shown a mean of 6.96 l/s.

Fig. 3. One-year solar irradiance profile

Fig. 4. Mean annual wind speed

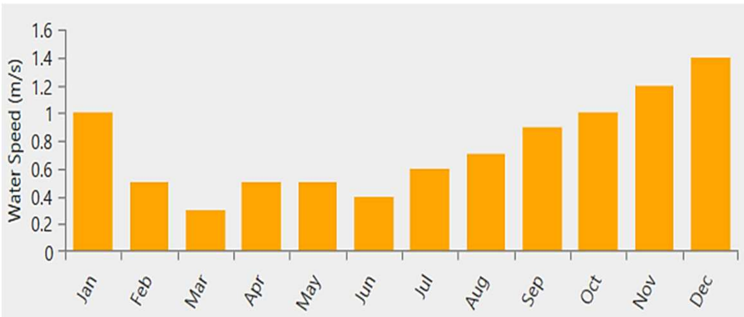

Fig. 5. Mean annual water speed

Fig. 6. Mean monthly flow rate

D. The island's load profile

The Figures below show the annual and island-specific electricity consumption curves. This load profile represents residential electricity consumption.

Our proposal for a hybrid system has been developed on Djerba Island, Tunisia. The structure of the hybrid system has been designed for 165.44 kWh/day in Fig.7.

Fig. 8 shows the island's load profile throughout the year, indicating that demand varies from month to month and time of day. In contrast, the peak power demand mainly takes place in the daytime and the evening. In particular, total power demand increases in hot weather and at low temperatures. The island's average monthly load profile is not uniform throughout the year. Peak demand occurs in June, July, and also in January.

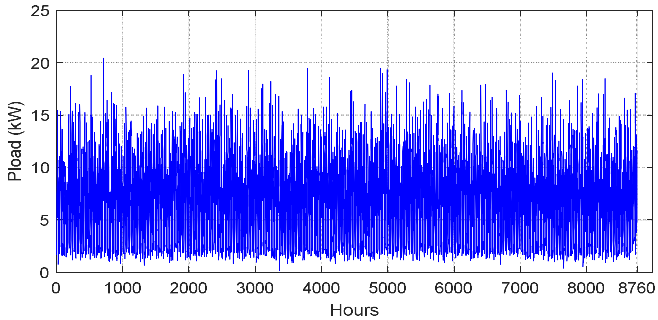

Fig. 7. Electricity consumption over one year

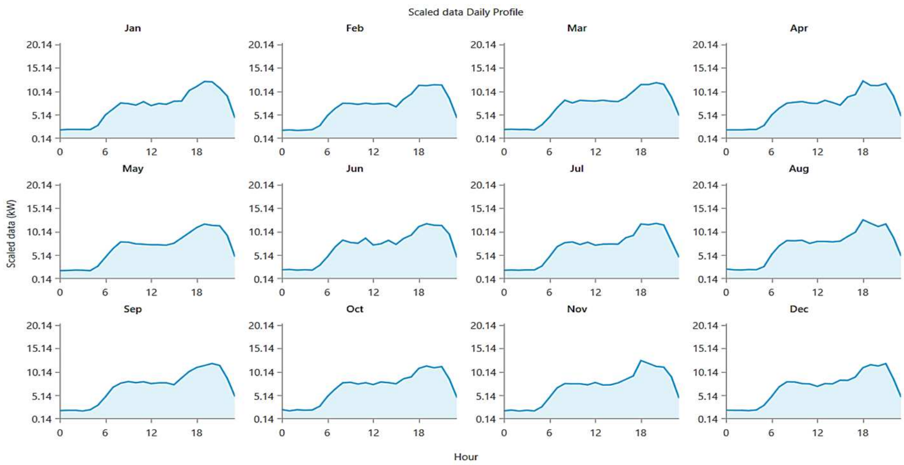

Fig. 8. Load profile of Island

III. ANALYSIS OF THE ECONOMIC SITUATION

Conventional generation processes generate significant operating and maintenance costs but have relatively low initial capital costs. On the other hand, renewable energy sources require high initial investments but have reduced operational and maintenance costs. In addition, a consideration of by taking into account a variety of environmental and economic variables, the software offers precious information on the dimensioning of power provision components and equipment [5]. The main economics calculated with the software in this context are total discounted cost (TDC) and the present value of energy (LCOE).

$$f_1 = \min(\mathrm{NPC}) \tag{1}$$

$$f_2 = \min(\mathrm{LCOE}) \tag{2}$$

The formula determined the net present cost below:

$$\mathrm{NPC} = \frac{C_{\mathrm{annual,total}}}{\mathrm{CPF}(j, R_{\mathrm{project}})} \tag{3}$$

with $C_{\text{annuel, total}}$ presents the sum annual cost, j is the annual effective rate of interest, R_{project} presents the period of life, and CPF is the coefficient of capitalization.

The present cost of a particular energy product is determined by the following equation (4):

$$\mathrm{COE} = \frac{C_{\mathrm{ann.t}}}{E_{\mathrm{served}}} \tag{4}$$

with $C_{\text{ann.t}}$ standing for total project-related annual expenditure, E_{served} denotes the amount of power generated in kWh (kilowatt-hours) over the year.

A. A factor of renewable energy

Our system is designed to reduce decentralized generation (DG), the net cost of generation (NPC) and the cost of energy (COE), thereby lowering CO2 emissions, and enhancing the system's environmental sustainability. The assessment is performed utilizing a measurement available at the renewable fraction (RF).

$$f_3 = \max\,(\mathrm{RF}) \tag{5}$$

A renewable factor determined by equation (6) in the following equation:

$$\mathrm{RF} = \left(1 - \frac{\sum P_{\mathrm{DG}}}{\sum P_{\mathrm{gen}}}\right) \times 100 \tag{6}$$

with P_{gen} represents the generated power of our system, P_{DG} represents the power produced by DG.

B. CO2 emissions

Distributed production presents several particular challenges, especially in terms of the impact on the environment. Hybrid systems' environmental performance is evaluated by assessing their CO2 content in the manufacturing process. LCE (Life Cycle Emissions) comprise both the energy consumed in manufacturing, transporting, and installing the components of the distributed generation system, and the carbon dioxide emissions resulting from the combustion of fuel in the system. The equation (7) is the basis for the distributed generation formula [5].

$$\mathrm{LCE} = \sum_{i=1}^{N} \beta_i E_L \quad (7)$$

Equation (7) gives the emissions of carbon dioxide over the generator's lifespan and the power produced or saved in a battery, by i (kg CO2-eq/kWh) and E_L (kWh).

C. Limits

Our variables are the photovoltaic power capacity (C_{pv}), the distributed generation capacity (C_{DG}), the hydraulic power capacity (C_{hy}), the total number of generators (N_{wt}), the total number of tidal generators (N_t), and the total number of battery packs (N_{bat}).

D. The decision factors

Decision variable limits are defined on the basis of the specific nature of each issue.

$$\begin{cases} 0 \le C_{pv} \le C_{pv}^{max} \\ 0 \le C_{DG} \le C_{DG}^{max} \\ 0 \le C_{hy} \le C_{hy}^{max} \\ 0 \le N_{wt} \le N_{wt}^{max} \\ 0 \le N_t \le N_t^{max} \\ 0 \le N_{bat} \le N_{bat}^{max} \end{cases} \quad (8)$$

E. Battery storage restrictions

Battery energy is regulated within the specified limits:

$$SOC_{min} \le SOC \le SOC_{max} \quad (9)$$

with SOC as charge state, and SOC_{min} and SOC_{max} as minimum and maximum production output, respectively.

IV. RESULTS AND DISCUSSION

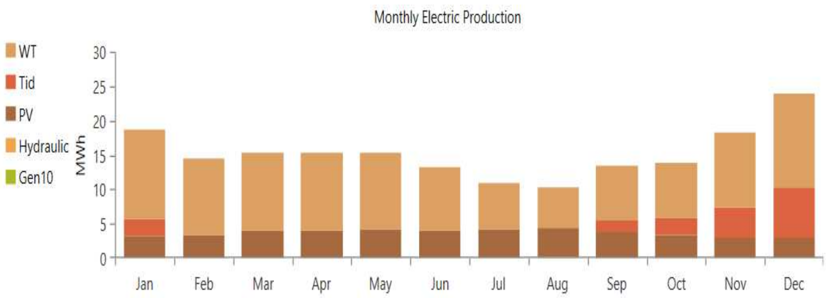

Fig. 9. Electric production

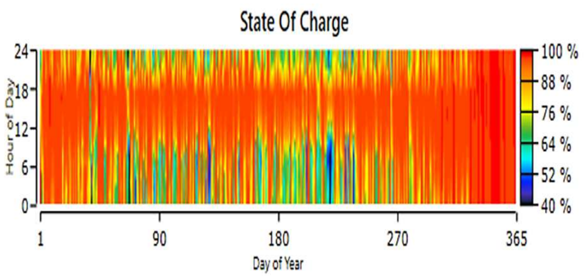

Fig. 10. Storage station state of charge

The monthly power generation of the optimal power system is illustrated in Fig. 9. The system's total annual electricity production by energy source represents 64.9% of the needs of the community at the studied site. Wind turbines account for 65.7%, while solar panels account for 23.9%, tidal power accounts for 10.1% of the total, and diesel generators for 0.293%. Hydroelectric installations and diesel generators on the island are not needed. Power generation using conventional renewable energy sources was identified as greater during winter than in spring and autumn.

The mean monthly power produced for every hybrid system component is shown in Table. II.

Following the implementation of a HERS, differences in environmental pollutants like carbon dioxide, carbon monoxide, and sulfur dioxide are significantly lower, as illustrated in Table III.

The battery bank's state of charge (SOC) distribution is presented in Fig. 10. The SOC allocation of the hybrid system is often close to the extremes. A SOC between 98% and 100% was achieved 43.55% of the time.

Fig. 11 shows that the initial investment cost (capital) is the highest, reaching 197,666.27$ for the total system. Operating and replacement costs are also significant, but lower. Salvage is slightly negative, while fuel costs are the lowest. Thus, capital costs dominate total expenditure, followed by operating and replacement costs, with relatively low salvage and fuel costs.

From Fig. 12, we can see that the battery fleet accounts for the largest share of the total net cost, amounting to 100, 364.94$. The total cost of diesel power generation is 93,446.92$. Photovoltaic panels cost 42,724.51$, as do wind turbines. Lastly, our lowest-cost constituent is the hydraulic system, with a total net cost of 2,292.75$.

TABLE II. POWER GENERATED BY THE VARIOUS COMPONENTS

Production	**kWh/yr**	**%**
PV system	43.829	23.9
Diesel generator	537	0.293
Wind turbine	120.204	65.7
Tudal turbine	18.471	10.1
Total	183.042	100

TABLE III. EMISSION OF THE VALUES

Type	Quantity	Unit
Carbon Dioxide	617	kg/year
Carbon Monoxide	4.67	kg/year
Unburned Hydrocarbons	0.170	kg/year
Particulate Matter	0.283	kg/year
Sulfur Dioxide	1.51	kg/year
Nitrogen Oxides	5.30	kg/year

Fig. 11. Distribution of cash flows according to type of cost

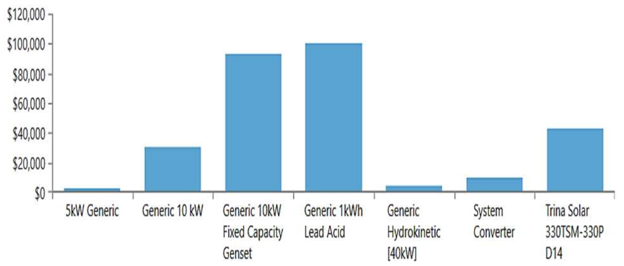

Fig. 12. Net costs by component

V. CONCLUSION

Our article proposes an optimal structural configuration to provide the community of Djerba, Tunisia with an appropriate component capacity, taking into account reduced costs, and lower CO2 emissions. At the end of the simulation, the systems were obtained and compared in terms of cost, energy production, and emissions. The principal conclusion to be taken is that:

- Net present cost (NPC) and energy cost (ECO) reductions were respectively $240.282 and $0.308/kWh.

- In practical emissions terms, the optimum system demonstrated considerable reduction of CO2, sulfur dioxide (SO2) as well as NOx, resulting in a reduction of 617 kg/year of CO2, 1.51 kg/year of SO2, and 5.30 kg/year of NOx.

Such results point to the necessity of predicting both energy and load patterns to provide better, more profitable, and more environmentally acceptable solutions monitoring concepts.

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